MILITARY SPECIFICATION
CHRONOGRAPH, RADAR SET: M36 WITH EQUIPMENT

1. SCOPE

1.1 This specification covers the M36 Radar Chronograph Set, hereinafter referred to as Chronograph. The Chronograph is a continuous wave doppler radar system with facilities for detecting a shift in the frequency between the transmitted signal and the received signal. This frequency shift (doppler effect) is a measure of the relative velocity of the projectile target with respect to the radar antenna.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-L-10470	Lights, Instrument, Fire Control Equipment
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-14811	Indicator, Digital, Display: 8215501
MIL-C-14817	Control, Frequency: 10525510
MIL-P-14818	Power Supply: 10512499
MIL-A-14819	Amplifier, Audio, Frequency: 10512508
MIL-A-14822	Amplifier, Detector: 10525674
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18 Manufacturing Processes and Inspection of
MIL-M-60817	Modules, Radar Chronograph: M36 (encapsulated)
MIL-T-60819	Telescope Straight, XM128
MIL-C-60228	Container, Shipping, Reusable 8566360
MIL-C-60230	Container, Shipping, Reusable for Radar Chronograph M36 Accessories:8566361

FSC 1290

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MIL-M-60673	Modules, Radar Chronograph: M36
MIL-C-60690	Chronograph Automatic Reliability Rater:10552370

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10525300	Radar Chronograph, Ass'y
F10512507	Mount, Radar Chronograph
C8213856	Tripod, Assembly
C8566401	Cable, Assembly, Special Purpose,Electrical
B8213855	Cable and Reel Assembly
D10559174	Cable Assembly, Special Purpose, Electrical
F8215513	Bracket Installation - (Jeep Mounting)
C10525559	Receiver, Radar Test
C10531933	Microphone Ass'y
F8205096	Telescope, Straight: XM128
F7654351	Case, Carrying M87
F10553463	Light, Instrument, M53E1
B7641600	Bracket, Lamp Assembly

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A5800940	Kit, Repair Parts
A8566360	Container, Shipping, Reusable
A8566361	Container, Shipping, Reusable (Accessories)
F10552370	Chronograph Automatic Reliability Rater
C11727583	Clamp Assembly (Stored W/Instrument Light)

PACKAGING DATA SHEET

10525235	Packaging of Radar Chronograph, M36

(Copies of specifications, standards, drawings and packaging data sheets required by Suppliers in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Fabrication. - The Chronograph shall be manufactured in accordance with the applicable drawings (see 2.1) pertaining thereto.

3.2 General Specifications. - The Chronograph shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship
e. Soldering electrical connections

3.2.2 MIL-C-46357

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes
e. Etched circuit assemblies

3.3 Subassemblies and Components. - Each and every Chronograph submitted in accordance with this specification shall incorporate subassemblies and components which have complied with all their respective drawing requirements, specification requirements (see 2.1) prior to incorporation into the Chronograph.

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3.4 Environmental. -

3.4.1 Shock. - The Chronograph shall be capable of three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

a. Bottom down
b. Front down
c. Right side down
d. Back down
e. Left side down
f. Top down

During the above shock, the Chronograph shall have Primary Input Power applied. The Digital Indicator shall have the Reset Light continuously lighted and the Nixie Tube Display on the Digital Counter shall continuously display all zeros and only zeros at all times. During and subsequent to the above shock, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.4.2 Vibration - The Chronograph shall be capable of being vibrated in a simple harmonic motion having an amplitude of 0.0075 (0.015 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz. The entire frequency range, from 10 to 55 hertz and return to 10 hertz, shall be uniformly traversed in one (1) minute ± 15 seconds. This motion shall be applied for a period of thirty (30) minutes ± 1 minute in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 1.5 hours. During the above vibration, the Chronograph shall meet the requirements of 3.6.4, 3.6.6, 3.6.13 and 3.6.14. At all other times during the vibration, the Chronograph shall have Primary Input Power applied. The Digital Indicator shall have the Reset Light continuously lighted and the Nixie Tube Display on the Digital Counter shall continuously display all zeros and only zeros at all times. During and subsequent to the above shock, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.4.3 Humidity. - The Chronograph shall be capable of continuous operation during four (4) hours ± 10 minutes of continuous exposure to a relative humidity of ninety (90) to ninety-five (95) percent at a temperature of plus 100 ± 5°F. During the above humidity, the Chronograph shall meet the requirements of 3.6 excluding 3.6.8.1, 3.6.9.1 and 3.6.10. During and subsequent to the above humidity, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.4.4 Sand and Dust. - The Chronograph shall be capable of withstanding three (3) hours of continuous exposure to blowing sand of particle size 0.01 to 1.00mm and blowing dust of particle size 0.0001 to 0.01mm with a density between 0.1 and 0.5 gram per cubic foot of dust at a temperature of plus 75°F ± 5°F, a relative humidity between fifteen (15) to thirty (30) percent, and a velocity of fifteen (15) knots. During and subsequent to the above sand and dust exposure, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

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3.4.5 Altitude

3.4.5.1 Operating. - The Chronograph shall be capable of continuous operation during four (4) hours of continuous exposure to an altitude of 10,000 feet at a temperature of plus 75°F ± 3°F. During the above altitude exposure, the Chronograph shall meet the requirements of 3.6 excluding 3.6.8.1, 3.6.9.1 and 3.6.10. During and subsequent to the above altitude exposure, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.4.5.2 Non-operating - The Chronograph shall be capable of withstanding four (4) hours of continuous exposure to an altitude of 50,000 feet at a temperature of minus 65°F ± 3°F. During and subsequent to the above altitude exposure, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.4.6 Fungus. - The Chronograph shall be capable of withstanding twenty-eight (28) days exposure to a composite spore suspension consisting of the following cultures:

a. Chaetomium globosom
b. Aspergillus flavus
c. Aspergillus versicalor
d. Penicillum luteum
e. Phizopus nigricans

The composite spore suspension shall be incubated at an ambient temperature of 81°F ± 4°F with an RH of 87 ± 2%. During and subsequent to the above exposure, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.4.7 Storage Temperatures - The Chronograph shall be capable of withstanding temperature stabilization at plus 160°F and minus 80°F for a period of eight (8) hours at each temperature extreme. During and subsequent to the above temperature extremes (160°F and -80F) the Chronograph shall not have been damaged or adversely affected and shall subsequently,after temperature stabilization at 75°F ± 15°F, meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.4.8 Operating Temperatures. - The Chronograph shall be capable of continuous operation for eight (8) hours at plus 125°F with 360 BTU/sq. foot/hour with a wind velocity of 6 knots. The Chronograph shall be capable of continuous operation for eight (8) hours at minus 50°F. During the above temperature extremes (plus 125°F with BTU/sq. foot/hour with a 6 knot wind velocity and minus 50°F), the Chronograph shall meet the requirements of 3.6 excluding 3.6.8.1, 3.6.9.1 and 3.6.10 at least once at the beginning, during the middle, and at the ending of each eight (8) hour temperature extreme. There shall be a minimum of five (5) minutes warm-up prior to performing the first tests. During and subsequent to the above temperature extremes, the Chronograph shall not be damaged or adversely affected and shall subsequently meet the requirements of 3.6 after each temperature extreme excluding 3.6.8.1 and 3.6.9.1.

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3.4.9 Rain - The Chronograph shall be capable of withstanding five (5) hours of continuous exposure to heavy rainfall at the rate of 2.0 inches/hour with a wind velocity of 35 knots at a water temperature of plus 85°F. During and subsequent to the above rainfall, the Chronograph shall not be damaged or adversely affected and shall subsequently shall no evidence of water penetration and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1

3.4.10 Salt Spray. - The Chronograph shall be capable of withstanding forty-eight (48) hours of continuous exposure to a fine spray of 5 % salt solution concentration with a specific gravity of from 1.0268 to 1.0413, a PH between 6.5 and 7.2 at a temperature between plus 93°F and plus 97°F. During and subsequent to the above salt spray, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.5 Electrical

3.5.1 Continuity - The Chronograph shall demonstrate continuity in accordance with all the wiring requirements as delineated on drawings constituting part of the Chronograph.

3.6 Performance

3.6.1 Performance Acceptance - The performance requirements specified herein shall be met on every Chronograph at Standard Ambient Conditions prior to Government Acceptance.

3.6.2 Primary Input Power. - The requirements of paragraph 3.6 shall be capable of being met over a Primary Power Input voltage range of 120 ± 5 VAC and a frequency range of 400 ± 20 Hz and maximum total distortion less than 3%.

3.6.3 Monitor Meters. - The Front Panel Monitor meter and the Amplifier-Detector Monitor meter shall indicate as shown in Table I for the appropriate Switch positions of the respective monitor meters.

TABLE I

Front Panel Monitor Meter and Switch

Switch Position	Monitor Meter Reading
Line	Green Band
B+	Green Band (Red Line)
Leakage	Green Band (Red Line)
Signal	Less than 10
AGCV	Green Band

Amplifier - Detector Monitor Meter and Switch

Switch Position	Monitor Meter Reading
AFC Disc	0 to ± 2 ua
Mod Drive	40 to 70 ua
Mod. Drive (-50°F)	30 to 55 ua
Mod Xtal 1	+45 to +90 ua
Mod Xtal 2	-45 to -90 ua
Mix Xtal 2	+20 to +80 ua
Mix Xtal 1	-20 to -80 ua

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3.6.4 Blower Motors. - Internal blower motors shall start immediately upon the application of Primary Input Power and the air flows shall be directed toward the Klystron.

3.6.5 Internal Calibration Signal. - The 75 KHz Internal Calibration oscillator shall be capable of producing an output on the Audio Amplifier (Audio Output receptical TP A3J1). The Output shall be an undistorted sine wave having a frequency of 75 KHz ± .012% and an amplitude of 600 ± 300 millivolts peak-to-peak. This Output signal shall be present at TP A3J1 with the O P E R A T E Switch on the Digital Indicator in both the CAL 1 and CAL 2 positions with the Velocity Selector Switch on the Audio Amplifier in the 925-1150 CAL position.

3.6.6 Receiver Sensitivity. - The Receiver sensitivity shall be such, that with a 45 MHz ± 0.01% + Δ F (Δ F varies over the Modulation Frequency range of 12 KHz to 120 KHz) undistorted sine wave injected into the Amplifier Detector at the Voltage Input levels and over the Modulation Frequency ranges as given in Table II; the Audio Output voltages and Audio Output Frequency range (TP A3J1) shall be an undistorted sine wave for each respective Velocity Selector switch position as given in Table II. At each Audio Output Frequency (end limit values), a Digital Indicator Nixie Tube display shall be obtained as given in Table II and shall be repeatable three times.

3.6.7 Leakage Tuning. - The Leakage adjustment controls (Leakage 1 and Leakage 2) shall be capable of being adjusted, turning smoothly without binding, to produce a Front Panel Monitor meter reading between +65 and +75 (Green Band) with the Front Panel Monitor Selector switch in Leakage Position. When tuning the Leakage controls, the Front Panel Monitor meter shall move in a smooth motion and direction consistent with the movement of the controls. With the above requirement accomplished, the Front Panel Monitor Selector switch in Signal position shall read less than +10 on the Front Panel Monitor meter for all positions of the Velocity Selector switch on the Audio Amplifier and the Audio Amplifier (TP A3J1) shall have a total voltage output of less than 50 millivolts peak-to-peak.

3.6.8 Transmitter Frequency and Power Output. -

3.6.8.1 The power output measured at the wave guide output with the Antenna Feed and Reflector Dish removed shall be a minimum of 250 milliwatts RMS at a frequency of 10,500 ± 5 MHz.

3.6.8.2 The radiated power output with the Antenna Feed and Reflector Dish connected shall be a minimum of 200 milliwatts RMS at a frequency of 10,500 ± 5 MHz.

3.6.9 System Sensitivity

3.6.9.1 With the Antenna Feed and Reflector Dish removed, the Chronograph shall produce a Digital Display Indicator count accurate to 0.1 per-cent or ± 2 counts, whichever is larger, for a Simulated Doppler Return signal of minus 80 decibels referred to one milliwatt (minus 80 DBM) over the Modulation Frequency Range of 12 KHz to 120 KHz with a transmitted frequency of 10,500 ± 5 MHz.

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3.6.9.2 With the Antenna Feed and Reflector Dish connected, the Chronograph shall produce a Digital Display Indicator count accurate to 0.1 per-cent or ± 2 counts, whichever is larger, for a Simulated Doppler Return signal of minus 80 decibels referred to one milliwatt (minus 80 DBM) over the Modulation Frequency Range of 12 KHz to 120 KHz with a radiated frequency of 10,500 ± 5 MHz.

TABLE II

Vel. Sel SW Pos.	Mod. Freq. Range KHz	Volt Input Level Micro-Volt (RMS)	Audio Out Frequency Range KHz	Audio Out Voltage MV P-P	Indicator Nixie Tube Display
70-325	12-23	7-500	12-23	100-1000	171.3-328.3
300-400	21-29	7-500	21-29	100-1000	299.7-413.7
375-525	26.3-37	5-500	26.3-37	125-1000	375.3-528.0
500-700	34-50	5-500	34-50	125-1000	485.3-713.7
650-800	45-57	5-500	45-57	125-1000	6420-813.3
750-1000	52-71	5-500	52-71	125-1000	742.0-1013.3
925-1150 CAL	63-81	5-500	63-81	125-1000	899.3-1156.0
1100-1400	76-98	5-500	76-98	125-1000	1084.7-1398.3
1400-1590	100-113	5-500	100-113	125-1000	1427.3-1612.7
1525-1800	106-120	5-500	106-120	125-1000	1512.7-1712.7

3.6.10. Antenna Pattern. - The Antenna beam-width between the 3 db points in both the E and H planes shall be 3.3° ± 0.3° over the frequency range of 10,495 MHz to 10,505 MHz. Over the frequency range of 10,495 MHz to 10,505 MHz, the maximum side lobes in both the E and H planes shall be down a minimum of 23 db and the cross-polarization components shall be down a minimum of 25 db. The Antenna gain shall be a minimum of 34 db over the frequency range of 10,495 MHz to 10,505 MHz.

3.6.11. Boresight. - The electrical (beam) and optical (telescope) axis of the Chronograph shall be parallel in both the vertical and horizontal planes within ± 3 mils and shall indicate the beam width in E and H planes.

3.6.12 CARR Simulator. - The Chronograph in conjunction with the CARR Simulator shall be capable of providing a system test of the functional reliability of the Chronograph.

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3.6.13 Elasped Time Meter. - The Elasped Time meter shall be capable of indicating the total time that Primary Power (400 ± 20 KHz and 120 ± 5 VAC) is applied within an accuracy of ± 10%.

3.6.14 Microphone. - With the Microphone connected by the Microphone Cable to the MICROPHONE Input on the Chronograph, the Reset Light shall go out when the Microphone is lightly tapped. This shall be accomplished with the MICROPHONE SENSITIVITY switch on the Digital Indicator over the range of 5 to 10.

3.6.14.1 Repeat the sequence of paragraph 3.6.14 while holding the OPER switch on the Digital Counter in the CAL 1 position. The Nixie Tube Display on the Digital Counter shall indicate a readout of 1070.7 ± 0.4. The Velocity Selector switch on the Audio Amplifier shall be in the 925-1150 CAL position.

3.6.15 Internal Voltage Test Points. - The voltages at TB1 and TB2 shall be as shown in Table III.

TABLE III

From	To	
TB1-6	TB1-2	+178 ± 4.0 VDC
TB1-5	TB1-2	+22 ± 0.5 VDC
TB1-4	TB1-3	+22 ± 0.5 VDC
TB2-6	TB2-3	-500 ± 10 VDC
TB2-5	TB2-3	-500 ± 10 VDC
TB2-6	TB2-5	6.6 ± 0.3 VAC
TB2-4	TB2-3	-500 ± 10 VDC (DELAYED)
TB2-2	TB2-3	-810 ± 60 VDC

3.6.16 Azimuth and Elevation Controls - The Chronograph shall be capable of being leveled by proper adjustment of the azimuth and elevation level adjust controls to within ± 2 mils concurrently in both the horizontal and vertical planes. The vial levels shall be capable of indicating the above level. Upon accomplishment of the above requirement, the Chronograph without further adjustment shall be capable of rotating 360° in the horizontal plane and meet the above requirement throughout the rotation with the single exception that the vial level on the mount (not the yoke) need not indicate a level condition within ± 2 mils.

3.6.17 Azimuth Ring. - The azimuth ring when set at a given deflection shall remain at the set deflection when the Yoke containing the Chronograph is rotated 360° clockwise and counter clockwise in the horizontal plane.

3.6.18 Automatic Frequency Control (AFC). - With P1 removed the AFC shall be capable of electronically tuning the Klystron over the frequency range of 10,480 to 10,520 MHz. The differential amplifier balance measured between test points J1 and J2 shall be capable of being adjusted to ± 0.25 VDC and within fifteen (15) minutes after adjustment shall not have drifted more than ± 3.0 VDC. The voltage between J3 and Ground shall be capable of being adjusted from -750 VDC to -870 VDC. When the Automatic Frequency Control (AFC) is electronically tuning the Klystron, the Amplifier Detector Monitor meter shall be capable of indicating control of the Klystron by deflection from 0 to +100

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(Center to right side) back to 0 (right side back to center) then from 0 to -100 (center to left side) back to 0 (left side back to center).

3.7 Reliability Run-in. - In order to maximize component reliability, prior to acceptance by the Government, each and every Chronograph shall be subject to the following Run-in conditions:

3.7.1 Temperature Run-in. - The Chronograph, without Primary Input Power applied, shall be temperature stabilized at plus 125°F and minus 25°F. At each temperature extreme, after temperature stabilization, Primary Input Power shall be applied and a five (5) warm-up interval shall exist prior to the commencement of testing. The Chronograph shall be temperature Run-in for a period of three(3) hours at each temperature extreme. During this three (3) hour test at each temperature extreme, the Chronograph shall be tested once to all the requirements of 3.6 excluding 3.6.8.1, 3.6.9.1 and 3.6.10. At all other times during a temperature extreme, the Chronograph shall have Primary Input Power applied. The Digital Indicator shall have the Reset Light continuously lighted and the Nixie Tube Display on the Digital Counter shall continuously display all zeros and only zeros at all times.

During and subsequent to the above temperature extremes, the Chronograph shall not be damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.7.2 Vibration Run-in - The Chronograph shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.0075 (0.0015 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz. The entire frequency range, from 10 to 55 hertz and return to 10 hertz shall be uniformly traversed in one (1) minute ± 15 seconds. This motion shall be applied for a period of thirty (30) minutes ± 1 minute in each of three (3) mutually perpendicular directions (x, y, z, axis) for a total of 1.5 hours. During the above vibration, the Chronograph shall meet the requirements of 3.6.4, 3.6.6, 3.6.13, and 3.6.14. At all other times during the vibration, the Chronograph shall have Primary Input Power applied. The Digital Indicator shall have the Reset Light continuously lighted & shall display zeros and only zeros at all times. During and subsequent to the above shock, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1

3.7.3 Shock Run-in. - The Chronograph shall be capable of three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

a. Bottom down
b. Front down
c. Right side down
d. Back down
e. Left side down
f. Top down

During the above shock, the Chronograph shall have Primary Input Power applied. The Digital Indicator shall have the Reset Light continuously lighted and the

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Nixie Tube Display on the Digital Counter shall continuously display all zeros and only zeros at all times. During and subsequent to the above shock, the Chronograph shall not have been damaged or adversely affected and shall subsequently meet the requirements of 3.6 excluding 3.6.8.1 and 3.6.9.1.

3.7.4 Failure During Reliability Run-in. - Replacement of defective components and correction of workmanship defects may be made during and subsequent to Run-in under the following conditions:

(a) Any part, module, component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the Reliability Run-in environmental requirements of the Chronograph.

(b) For any failure occurring during a temperature extreme, any replacement part shall be exposed for at least one (1) hour at the extreme temperature at which the failure occurred prior to being utilized as a replacement during an extreme temperature.

(c) After a correction has been accomplished during an extreme temperature, at least 1.5 hours of operation with a minimum of one (1) complete test sequence to the requirements of 3.6 excluding 3.6.8.1, 3.6.9.1, and 3.6.10 shall be sucessfully accomplished at the temperature extreme at which the failure occurred.

(d) Any failure occurring during 3.7.2 or 3.7.3 shall necessitate a complete re-run of the requirements of 3.7.2 and 3.7.3.

(e) Any failure occurring subsequent to a Run-in condition (para 3.7.1, 3.7.2, 3.7.3) shall be considered to have occurred during the Run-in condition and shall necessitate a satisfactory re-run of the Run-in condition at which the failure is assumed to have occurred.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the Contract or Purchase Order, the Supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the Supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First Article--Testing and Approval - The requirements for First Article Approval and the responsibility (Government or Contractor) for First Article Testing shall be as specified in the Contract. The sample for First Article Approval Tests shall consists of three (3) Chronographs manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government Testing - When the Government is responsible for conducting First Article Testing (as specifically delineated in the Contract), the Contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the Contract and submit a record of this inspection with the sample, including Certificates of Conformance for materials.

4.2.2 Contractor Testing - When the Contractor is responsible for conducting First Article Testing (as per para 4.2), the sample shall be inspected by the Contractor for all the requirements of the Contract. The sample and a record of this inspection, including Certificates of Conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the Contractor's inspection.

4.3 Inspection Provisions.

4.3.1 Examination and Test.

4.3.2 Acceptance and Rejection - Rejected Lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected Lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling- Subsequent to First Article Approval (see 4.2), three (3) units shall be selected at random by a Government representative from the first fifty (50) units, or portion thereof, produced and two(2) additional units from each subsequent fifty (50), or portion thereof, produced. The Special Sample units shall have successfully met all examinations and acceptance tests specified herein and no Lots from which the Special Sample units have been selected shall be accepted by the Government until the Special Sample units have successfully met the following requirements.

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
401.	Fabrication	3.1	Applicable Dwg & Specs.
402.	General Specifications	3.2	MIL-F-13926 & MIL-C-46357
403.	Shock	3.4.1	4.6.1.2
404.	Vibration	3.4.2	4.6.1.3
405.	Humidity	3.4.3	4.6.1.4
406.	Sand and Dust	3.4.4	4.6.1.5
407.	Altitude-operating	3.4.5.1	4.6.1.6.1
408.	Altitude-non-operating	3.4.5.2	4.6.1.6.2

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CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
409. Storage temperatures	3.4.7	4.6.1.8.1
410. Operating temperatures	3.4.8	4.6.1.8.2
411. Rain	3.4.9	4.6.1.9
412. Salt Spray	3.4.10	4.6.1.10

4.4.1 Failure of Special Sample - Should any one unit of a Special Sampling (see 4.4) fail to meet the specified test requirements, necessary corrections shall be made by the Contractor to the Lot of units from which the Special Samples were selected and reselected Special Samples per paragraph 4.4 shall be submitted and tested for approval.

4.5 Inspection equipment - Except as otherwise provided for by the Contract, the Contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government Furnished Inspection Equipment. - Where the Contract provides for Government-Furnished Test Equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor Furnished Inspection Equipment.

4.5.2.1 Contractor Design. - The Contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of product which does not exceed 90% of the prescribed tolerance limit and permit positive rejective when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test Methods Procedures.

4.6.1. Environmental. - Test equipment and test facilities utilized in the following environmental tests shall be in accordacne with "Test Facilities" requirements of MIL-F-31926 and conditions specified in 3.4.

4.6.1.1. Equipment Configuration. -

4.6.1.1.1 The equipment configuration for the Shock (para 4.6.1.2) and Vibration (para 4.6.1.3) shall be as follows:

(a) The Radar Chronograph Assembly (10525300) shall be functionally assembled to the Mount, Radar Chronograph (10512507).

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(b) The equipment configuration above (4.6.1.1.1) (a) shall be functionally assembled to the Tripod Assembly (8213856)

(c) The Telescope, XM128 (8205096) shall be functionally assembled to the Radar Chronograph Assembly (10525300) by the Mount, Telescope (10531919).

(d) The Primary Input Power cable (10531920) shall be functionally assembled to the Primary Input Power receptical (A6J1) of the Radar Chronograph Assembly (10525300)

(e) The Microphone cable (10531921) shall be functionally assembled to the Microphone Assembly (10531933) and to the Microphone Input receptical (A3P1) of the Radar Chronograph Assembly (10525300)

4.6.1.1.2 The equipment configuration for all other Environmental Tests excluding Shock (para 4.6.1.2) and Vibration (para 4.6.1.3) shall be as follows:

(a) The Radar Chronograph assembly (10525300) shall be functionally assembled to the Mount, Radar Chronograph (10512507).

(b) The equipment configuration above (4.6.1.1.1) (a) shall be functionally assembled to the Tripod Assembly (8213856).

(c) The Telescope, XM128 (8205096) shall be functionally assembled to the Radar Chronograph Assembly (10525300) by the Mount, Telescope(10531919).

(d) The Primary Input Power cable (10531920) shall be functionally assembled to the Primary Input Power receptical (A6J1) of the Radar Chronograph Assembly (10525300)

(e) The Microphone cable (10531921) shall be functionally assembled to the Microphone Assembly (10531933) and to the Microphone Input receptical (A3P1) of the Radar Chronograph Assembly (10525300).

(f) The Receiver, Radar Test (10525559) shall be functionally assembled to Cable Ass'y, Special Purpose, Electrical (8566401) and shall be exposed to the evnironment exisitng during the Environmental Tests.

(g) The Chronograph Automatic Reliability Rater (10552370) shall be removed from the Container Shipping, Reusable (ascessories) (8566361) and functionally assembled to the Radar Chronograph Assembly (10525300). The Chronograph Automatic Reliability Rater (10552370) shall have its lid opened.

(h) The Light, Instrument, M53E1 (10553464) shall be functionally assembled to Clamp Assmebly (11727583) and to Bracket, Lamp Assembly (7641600). The above shall then be functinnally assembled to the Mount, Radar Chronograph (10512507) and to Telescope, XM128 (8205096)

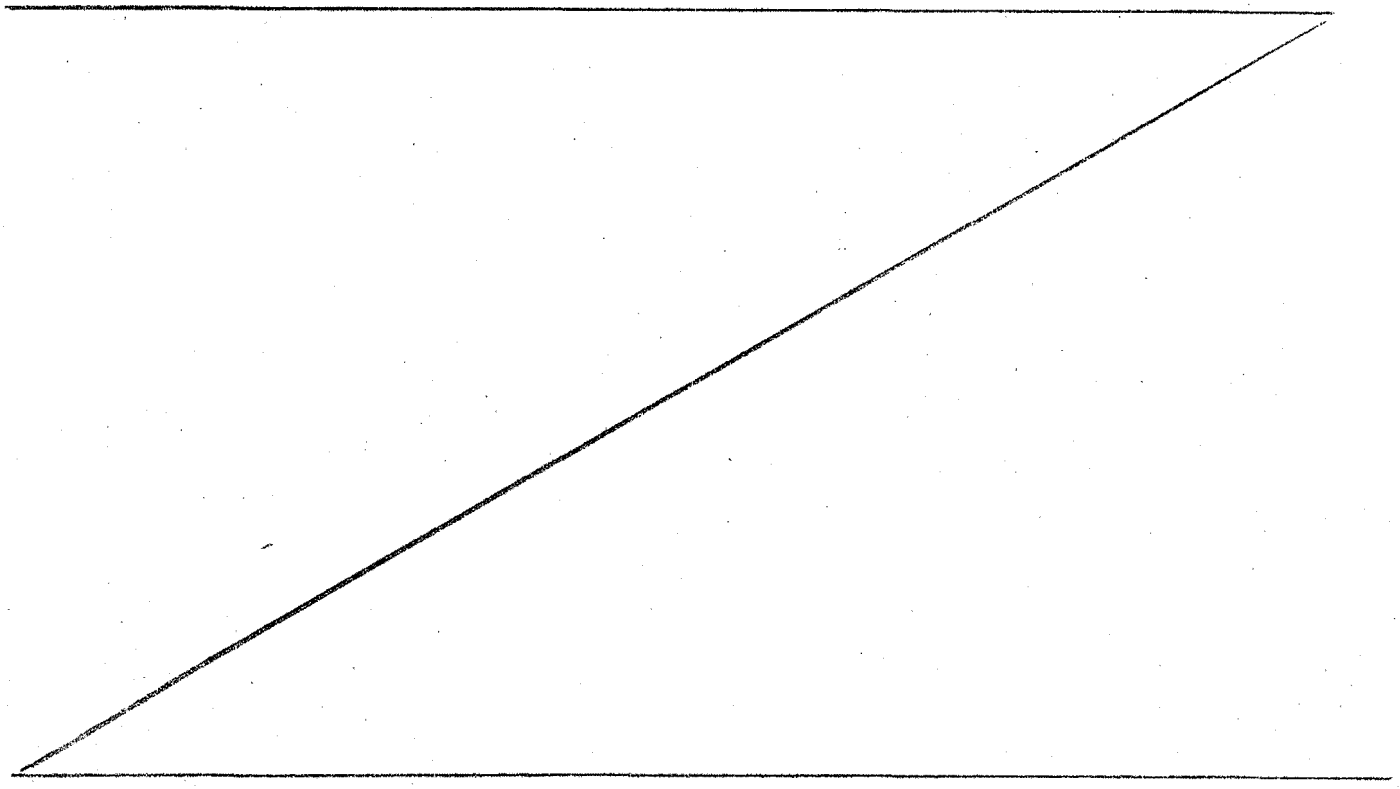
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(i) The Case, Carrying M87 (7654351) the Receiver, Radar Test (Accessory Case) (10525599), the Bracket Installation (Jeep Mounting) (8215513), the Container, Shipping, Reusable (8566360), the Container, Shipping, Reusable (acessories) (8566361), and the two Reels (10525702) and Mounting Plate (10525509) shall be exposed to the environment existing during the Environmental Tests in their functionally operational condition.

(j) The Cable Assembly, Special Purpose, Electrical (10559174) shall be removed from Case Carrying (10559180) and both shall be exposed to the environment existing during the Environmental Test.

4.6.1.2 Shock. - The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.4.1. Subsequent to the eighteen (18) shocks, the Chronograph shall be examined for damage, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.1.

4.6.1.3 Vibration - The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.4.2. During the above vibration, the Chronograph shall meet the performance test as specified in 3.4.2. Subsequent to the 1.5 hour vibration, the Chronograph shall be examined for damage, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.2.



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4.6.1.4 Humidity. - The humidity test shall be performed as specified in MIL-STD-810, Method 507, Procedure I for four (4) hours at the conditions specified in 3.4.3. Subsequent to this test, the Chronograph shall be examined for damage and adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.3.

4.6.1.5 Sand and Dust. The sand and dust test shall be performed as specified in MIL-STD-810, Method 510, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.4.4. Subsequent to this test, the Chronograph shall be examined for damage and adverse affects then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.4.

4.6.1.6 Altitude

4.6.1.6.1 Altitude Operating - The altitude test shall be performed as specified in MIL-F-31926, Procedure I at an altitude of 10,000 feet and as specified in 3.4.5.1. The Chronograph shall be exposed to the above condition for a time period required to allow all entrapped air in the chamber to escape as evidenced by chamber pressure stabilization. The barometric pressure shall then be maintained to simulate 10,000 feet (20.58 inches of mercury) altitude condition. While the Chronograph is maintained at this pressure for four (4) hours, the Chronograph shall be subjected to the tests specified in 3.4.5.1. Subsequent to this test, the Chronograph shall be examined for damage and adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.5.1.

4.6.1.6.2 Altitude Non-Operating - The Chronograph shall be subjected to barometric pressure specified in MIL-F-13926, Procedure II for four (4) hours as specified in 3.4.5.2. Subsequent to this test, the Chronograph shall be examined for damage and adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.5.2.

4.6.1.7 Fungus - One (1) Chronograph shall be subjected to the fungus test specified in MIL-F-13926 at the conditions specified in 3.4.6. Subsequent to the 28 day incubation period, the Chronograph shall be examined for damage and adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.6. Subsequent acceptance of the Chronograph shall be by Certification.

4.6.1.8 Temperature Cycling.

4.6.1.8.1 Storage Temperatures - This test shall be performed as specified in MIL-F-13926 Procedure II and shall be thermally stabilized at +160°F for eight (8) hours and at -80°F for eight (8) hours. The Chronograph shall be returned to room ambient temperature +75 ± 15°F, examined for damage and adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.7.

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4.6.1.8.2 Operating Temperatures - The Chronograph shall be placed in a test chamber and the temperature gradually raised at a rate not to exceed 2°F per minute until thermally stabilized at + 125°F. With 360 BTU/sq. foot/hour radiated vertically downward upon the Chronograph and a horizontal wind velocity of 6 knots to either side of the Chronograph, the Chronograph shall be tested as specified in 3.4.8. Subsequent to the above eight (8) hour temperature extreme (+125°F with 360 BUT/sq.foot/hour and a 6 knot wind), the Chronograph shall be examined for damage or adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.8. The Chronograph shall be placed in a test chamber and the temperature gradually lowered at a rate not to exceed 2°F per minute unitl thermally stabilized at -50°F. The Chronograph shall be tested as specified in 3.4.8. Subsequent to the above temperature extreme (-50°F), the Chronograph shall be examined for damage or adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.8.

4.6.1.9 Rain - The rain test shall be performed as specified in MIL-F-13926, Procedure I with the conditions as specified in 3.4.9. Subsequent to this test, the Chronograph shall be examined for damage and adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.9.

4.6.1.10 Salt Spray - The salt spray test shall be performed as specified in MIL-STD-202, Method 101, Condition B with the conditions as specified in 3.4.10. Subsequent to this test, the Chronograph shall be examined for damage and adverse affects, then tested as specified in 4.6.2 excluding 4.6.2.8.1 and 4.6.2.9.1 for compliance with 3.4.10.

4.6.2 Performance - Unless otherwise specified, the performance tests specified herein shall be satisfactorily accomplished on every Chronograph at Standard Ambient Conditions. Standard Ambient Conditions for the purposes of these tests shall be defined as:

(a) Temperature between 65°F and 85°F.

(b) Barometric pressure between 28 and 32 inches of mercury.

(c) Relative humidity (RH) up to 90%

4.6.2.1 Primary Input Power - Unless otherwise specified, the Primary Input Power applied to the Chronograph shall be as delineated in the respective test paragraphs of para 4.6.2

(a) 120 RMS volts, single phase, frequency 400 hertz, maximum total distortion less than 3% applied to A6J1-A. Neutral to A6J1-B and Ground A6J1-C.

(b) 115 RMS volts, single phase, frequency 380 hertz, maximum total distortion less than 3% applied to A6J1-A. Neutral to A6J1-B and Ground to A6J1-C.

(c) 125 RMS volts, single phase, frequency 420 hertz, maximum total distortion less than 3% applied to A6J1-A. Neutral to A6J1-B and Ground to A6J1-C.

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4.6.2.2 Blower Motors - Upon application of Primary Input Power per para 4.6.2.1 (c), the both Blowers shall immediately start and continue running with the air-flows directed toward the Klystron.

4.6.2.2.1 Repeat the tests of para 4.6.2.2 with an applied Primary Input Power per para 4.6.2.1 (b) and per para 4.6.2.1 (a).

4.6.2.3. Monitor Meters - Apply Primary Input Power per para 4.6.2.1 (a). The Blowers shall run and the red Pilot Light on the Power Supply shall be continuously lighted. Allow a five (5) minute warm-up interval, then properly adjust the Automatic Frequency Control per para 4.6.2.4.1 and the Leakage Tuning per para 4.6.2.5. The Front Panel Monitor meter and the Amplifier-Detector Monitor meter shall indicate for the respective Switch Positions the values as given in Table I.

4.6.2.3.1 Repeat the tests of para 4.6.2.3 with an applied Primary Input Power per para 4.6.2.1 (b) and per para 4.6.2.1 (c).

4.6.2.4. Automatic Frequency Control (AFC) - Apply Primary Input Power per para 4.6.2 (a). With the Amplifier-Detector Monitor meter in the AFC position, disconnect coaxial cable P1 from the Discriminator section of the wave guide. Adjust BAL potentiometer (R10) to obtain a voltage between test points J1 and J2 having a value of $\pm$ 0.25 VDC. The maximum drift in this voltage shall be as given in para 3.6.18. Adjust the AFC potentiometer (R17) fully clockwise. The voltage between J3 and Ground shall be more negative than -870 VDC. The Amplifier-Detector Monitor meter shall read zero. Slowly rotate AFC potentiometer (R17) counter-clockwise, while monitoring the radiated power from the Chronograph, until the Amplifier-Detector Monitor meter reads +100 (right side deflection). The frequency of the radiated power from the Chronograph shall have traversed from a frequency greater than 10,520 MHz to a frequency of 10,506 MHz $\pm$ 1 MHz at an Amplifier-Detector Monitor meter reading of +100. Continue slowly rotating AFC potentiometer (R17) counter-clockwise while monitoring the radiated power from the Chronograph until the Amplifier-Detector Monitor meter reads -100 (left side deflection). The radiated power from the Chronograph shall have traversed from a frequency of 10,506 MHz $\pm$ 1 MHz to 10,494 MHz $\pm$ 1 MHz. Continue slowing rotating AFC potentiometer (R17) counter-clockwise to the adjustment stop. The Amplifier-Detector Monitor meter shall read zero. The radiated power from the Chronograph shall have traversed from a frequency of 10,494 MHz $\pm$ 1 MHz to a frequency less than 10,480 MHz. The voltage between J3 and Ground shall be more positive than -750 VDC.

4.6.2.4.1 Repeat the tests of para 4.6.2.4 with an applied Primary Input Power per para 4.6.2.1 (b) and per para 4.6.2.1 (c).

4.6.2.4.2 Repeat the test procedure of para 4.6.2.4 except do not monitor the frequency of the radiated power from the Chronograph or the voltage between J3 and Ground. Then readjust AFC Potentiometer (R17) to the zero reading that exists between the positive and negative swing of the Amplifier-Detector Monitor meter. Reconnect coaxial cable J1 to the Discriminator section of the wave guide. The Amplifier-Detector Monitor meter shall read 0 $\pm$ 2 u a.

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4.6.2.5 Leakage Tuning - Connect an Oscilloscope to TP A3J1 of the Audio Amplifier. Place the Velocity Selector switch in the 70-325 position. Apply Primary Input Power per para 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor meter switch to the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2. Place the Front Panel Monitor selector switch in the LEAKAGE position. Adjust the Leakage adjustment controls meeting the requirements of para 3.6.7. Without further adjustment of the Leakage controls, rotate the Velocity Selector switch thru the remaining nine (9) positions. The Audio Output voltage (TP A3J1) and Front Panel Monitor meter shall meet the requirements of para 3.6.7 for each position of the Velocity Selector switch on the Audio Amplifier.

4.6.2.5.1 Repeat the tests of para 4.6.2.5 with an applied Primary Input Power per para 4.6.2.1 (b) and per para 4.6.2.1 (c).

4.6.2.6 Internal Voltage Test Points - Apply Primary Input Power per para 4.6.2.1 (a). Allow a one (1) minute warm-up interval. The voltages on TB1 and TB2 shall be as given in para 3.6.15 Table III.

4.6.2.7 Receiver Sensitivity - Disconnect Amplifier-Detector coaxial cables J3 and J4 from the Mixer section of the wave guide and connect the cables together using a coaxial tee. Connect the tee to a 45 MHz Signal Generator and to a Frequency Counter. Insert a coaxial tee connector between the J5 coaxial cable (Modulator Output) and the wave guide Modulation connector in such a manner that the Local 45 MHz Oscillator of the Amplifier-Detector may be monitored with a Frequency Counter. Connect a Frequency Counter and an Oscilloscope to TP A3J1 of the Audio Amplifier. Connect the Microphone to one end of the Microphone cable. Connect the other end of the Microphone cable to the MICROPHONE input on the Chronograph. Set the MICROPHONE SENSITIVITY adjustment on the Digital Counter to 10. Apply Primary Input Power per para 4.6.2.1 (b); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor switch to the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Place the VELOCITY Selector switch on the Audio Amplifier to the first position (70-325) as given in Table II. The Local 45 MHz Oscillator of the Amplifier-Detector shall have a frequency of 45 MHz $\pm$ 0.01 percent. The difference between the applied frequency from the signal Generator and the frequency of the Local 45 MHz Oscillator is the applied Modulation Frequency. Adjust and hold the Signal Generator for a voltage Input level of 7 microvolts (RMS) as given in Table II. Vary the Signal Generator frequency such that the Modulation Frequency range (12-23) as given in Table II is sweep. The Audio Output Frequency range (12-33) and Audio Output voltage (100-1000) shall be an undistorted sine wave with values as given in Table II. Adjust the Signal Generator frequency such that the Modulation Frequency end limit value of 12 KHz exists. Set the DELAY Selector switch on the Digital Counter to position 3. Reset the Digital Indicator. Lightly tap the Microphone. The Nixie Tube display (171.3) shall be as given in Table II. Reset and lightly tap the Microphone three times getting a Nixie Tube display (171.3) as given in Table II. Repeat the above tests at a Modulation Frequency end limit value of 23 KHz per Table II. The Nixie Tube display (328.3) shall be as given in Table II.

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Repeat the above tests at Modulation frequency end limit values (12 KHz and 23 KHz) with a Voltage Input Level of 500 microvolts (RMS) per Table II. Adjust The Signal Generator such that the Modulation Frequency is the mid-range value (17.5 KHz) of the Modulation Frequency Range as per Table II. Vary the Signal Generator Voltage Input level (7-500) as given in Table II. The Audio Output voltage (100-1000) per table II, shall be an undistorted sine wave having a frequency of 17.5 KHz.

4.6.2.7.1 Repeat the test procedure of para 4.6.2.7 for the remaining nine (9) positions of the Velocity Selector switch using the respective criteria as given in Table II.

4.6.2.8 Transmitter Frequency and Power Output

4.6.2.8.1 Antenna Feed and Reflector Dish removed - Remove the Antenna Feed and Reflector Dish. Connect in place of the Antenna Feed an X-band power Meter and Frequency Counter. Apply Primary Input Power per para 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor switch to the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2. The power output shall be a greater than 250 milliwatts RMS at a frequency of 10,500 $\pm$ 5 MHz.

4.6.2.8.2 Repeat the tests of para 4.6.2.8.1 with an applied Primary Input Power per para 4.6.2.1 (b) and per para 4.6.2.1 (c).

4.6.2.8.3 Antenna Feed and Reflector Dish connected - Assemble the Antenna Feed and the Reflector Dish to the Chronograph. This test shall be performed in an open area having no reflective objects capable of distorting the measurements. Apply, Primary Input Power per para 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor switch in the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2. Using a Calibrated Horn and Crystal Mount, place the Calibrated Horn at the Calibration distance and measure the crystal current. Referring to the Calibration Curve associated with the Calibrated Horn, calculate the total radiated power. The total radiated power shall be equal to or greater than 200 milliwatts RMS. Replace the Calibrated Horn with an X-band frequency meter. The frequency of the radiated power shall be 10,500 $\pm$ 5 MHz.

4.6.2.8.4 Repeat the tests of para 4.6.2.8.3 with an applied Primary Input Power per para 4.6.2.1 (b) and per para 4.6.2.1 (c).

4.6.2.9 System Sensitivity

4.6.2.9.1 Antenna Feed and Reflector Dish removed - Remove the Antenna Feed and Reflector Dish. In place of the Antenna Feed, connect test equipment capable of sampling the frequency of the transmitted power,

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lower shifting the frequency of the transmitted power over the range of 2 KHz to 120 KHz(Modulation Frequency Range), and reflecting the lowered shifted frequency to the Chronograph receiver at a minus 80 decibels level referred to one (1) milliwatt (minus 80 dbm). Connect the Microphone thru the Microphone Cable to the MICROPHONE input receptical on the Front Panel of the Chronograph. On the Digital Counter, place the Delay Selector to position three (3) and the MICROPHONE Sensitivity potentiometer to position five (5). Apply Primary Input power per para 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor switch in the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2. and the Leakage Tuning per para 4.6.2.5. As given in Table IV, with the Velocity Selector switch in the first position (70-325) and a minus 80 dbm reflected signal level at a Modulation Frequency as per column A (13 KHz), lightly tap the Microphone. The Nixie Tube Display on the Digital Counter shall be as given in column A, Table IV, with a count tolerance per para 3.6.9.1. Repeat the above procedure for the remaining nine (9) positions of the Velocity Selector switch; with Lower Shifted Frequency per column A and Digital Indicator Nixie Tube Display per column A, as given in Table IV.

4.6.2.9.1.1 Repeat the test procedure of para 4.6.2.9.1 using a Primary Input Power per para 4.6.2.1 (b) using column B values from Table IV.

4.6.2.9.1.2 Repeat the test procedure of para 4.6.2.9.1 using a Primary Input Power per para 4.6.2.1 (c) using column C values from Table IV.

4.6.2.9.2 Antenna Feed and Reflector Dish connected - Assemble the Antenna Feed and Reflector Dish to the Chronograph. This test shall be performed in an open area having no reflective objects capable of distorting the measurements. At a distance of 75 meters in front of the Chronograph place test equipment capable of sampling the frequency of the radiated power, lower shifting the frequency of the radiated power over the range of 12 KHz to 120 KHz (Modulation Frequency Range), and transmitting the lowered shifted frequency back to the Chronograph at a minus 80 decibels level referred to one (1) milliwatt (minue 80 dbm). Connect the Microphone thru the Microphone Cable to the MICROPHONE input receptical on the Front Panel of the Chronograph. On the Digital Counter, place the Delay Selector to position three (3) and the MICROPHONE Sensitivity potentiometer to position five (5). Apply Primary Input Power per para 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor switch to the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2 and the Leakage Tuning per para 4.6.2.5. As given in Table IV, with the Velocity Selector switch in the first position (70-325) and a minus 80 dbm reflected signal level at a Modulation Frequency as per column A (13 KHz), lightly tap the Microphone. The Nixie Tube Display on the Digital Counter shall be as given in column A, Table IV, with a count tolerance per para 3.6.9.2. Repeat the above procedure for the remaining nine (9) positions of the Velocity Selector switch; with Lower Shifted Frequency per column A and Digital Indicator Nixie Tube Display per column A, as given in Table IV.

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4.6.2.9.2.1 Repeat the test procedure of para 4.6.2.9.2 using a Primary Input Power per para 4.6.2.1 (b) using a column B values from Table IV.

4.6.2.9.2.2 Repeat the test procedure of para 4.6.2.9.2 using a Primary Input Power per para 4.6.2.1 (c) using column C values from Table IV.

TABLE IV

VEL. SEL. SW. POS.	MODULATION FREQUENCY KHz			DIGITAL INDICATOR NIXIE TUBE DISPLAY		
	(A)	(B)	(C)	(A)	(B)	(C)
70-325	13	17	21	185.3	242.7	299.7
300-400	22	24	28	311.0	342.7	399.7
375-525	29	32	36	414.0	456.7	514.0
500-700	38	44	49	542.3	628.3	699.7
650-800	47	51	56	671.0	728.3	799.3
750-1000	54	61	69	771.0	871.0	985.3
925-1150 CAL	67	74	70	956.7	1056.0	1128.0
1100-1400	78	85	96	1113.7	1213.7	1370.7
1400-1500	103	107	111	1470.7	1527.7	1585.0
1525-1800	109	114	119	1556.3	1627.7	1699.3

4.6.2.10 Antenna Pattern - These tests shall be performed in an open area having no reflective objects capable of distorting the measurements. A test range with equipment designed for antenna testing including suitable receiving and recording equipment shall be utilized. The tests shall be performed in a manner capable of positively demonstrating compliance with the requirements of para 3.6.10. In order to meet the requirements of para 3.6.10 over the frequency range of 10,495 MHz to 10,505 MHz, the Antenna Ass'y (10525626) shall be tested to the requirements of 3.6.10 at a frequency of 10,495 MHz, a frequency of 10,500 MHz, and a frequency of 10,505 MHz. This shall be accomplished prior to final assembly. Subsequent tests to the requirements of para 3.6.10 shall be performed at a frequency of 10,500 MHz $\pm$ 5 MHz with Primary Input Power as per para 4.6.2.1 (a).

4.6.2.11 Azimuth and Elevation Controls - Place the Tripod in an upright position. Extend the three Tripod legs such that one leg is extended 0.5 feet, one leg is extended 1.0 feet, and one leg is extended 1.5 feet. Place the Chronograph and Mount on top of the Tripod and seat the base of the Mount into the Leveling Base Adapter Ring. Rotate Chronograph until the bubble vial on the Mount indicates a level condition. Lock the Mount in place by tightening Leveling Base Locking handles. Loosen Azimuth Deflection Lock and rotate Chronograph until Deflection Pointer coincides with the middle of the white band. Loosen Elevation Leveling Lock handle and turn Elevation Adjustement Drive knob until the bubble vial on the Yoke indicates a level condition. Tighten the Elevation Leveling Lock Handle. Place a Level Vial capable of indicating a $\pm$ 2 mil out of level condition on the Telescope Mount. (The Telescope Mount must have been adjusted previously as per para 4.6.2.13) Loosen the Azimuth Deflection Lock and rotate the Chronograph 360 degrees clockwise and 360 counter-clockwise. The Level Vial on the Telescope Mount shall indicate a level condition with $\pm$ 2 mils. The bubble vial on the Yoke shall not shift more than $\pm$ one (1) vial division.

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4.6.2.12 Azimuth Ring - Set and lock the azimuth ring at 0, 800, 1600, 2400, and 3200 mils in both the clockwise and counter-clockwise directions. The azimuth ring shall remain at the initial setting when the Chronograph is rotated 360 degrees clockwise and 360 degrees counter-clockwise in horizontal plane.

4.6.2.13 Boresight - Secure the Telescope XM128 to the Telescope Mount. Place a Receiver Horn connected to an X-band Power Meter at a distance of 50 meters in front of the Chronograph. Apply Primary Input Power per para. 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2. Level the Chronograph within $\pm$ 2 mils in both the horizontal and vertical planes. Aim and secure the Chronograph for a maximum power reading (both vertically and horizontally). While sighting thru the Telescope XM128, adjust the rear elevation screws in the Telescope Mount until the horizontal cross-hair is ten (10) inches above center of the Receiver Horn and the vertical cross-hair is in the center of the Receiver Horn with a tolerance as given in paragraph 3.6.11.

4.6.2.13.1 Connect the Microphone Cable to the HORN Output of the CARR and connect the Radar Test Receiver to the other end of the Microphone Cable. Place the Radar Test Receiver a distance of 45 feet in front of the Chronograph and 4 feet above the ground. Place the Selector switch on the CARR in the METER position. Apply Primary Input Power per para. 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Adjust the Automatic frequency Control (AFC) per para. 4.6.2.4.2. Level the Chronograph within $\pm$ 2 mils in both the horizontal and vertical planes. Aim and secure the Chronograph for a maximum meter indication on the CARR meter (both vertically and horizontally). Rotate the Chronograph in the horizontal plane (horizontally left) until the CARR meter indicates .707 of the maximum meter indication. Sight thru the Telescope XM128 and record the deflection in mils. Repeat the above procedure in the horizontal plane (horizontally right). The sum of the left deflection and right deflection shall be 58.6 $\pm$ 5.3 mils (Beam width $3.3^{o} \pm 0.3^{o}$). Repeat the above procedure in the vertical plane the sum of the up deflection and down deflection shall be 58.6 $\pm$ 5.3 mils (Beam width $3.3^{o} \pm 0.3^{o}$).

4.6.2.13.2 Connect the Instrument Light to the Clamp Assembly and to the Bracket Lamp Assembly. Functionally assemble the above to the Mount and to the Telescope XM128. Adjust the brightness control on the Instrument Light and assure that the reticle (mil scale) in the Telescope XM128 is illuminated.

4.6.2.14 CARR Simulator - Secure the CARR Simulator to the right side of the Chronograph. Connect the Microphone Cable to the HORN Output of the CARR and connect the Radar Test Receiver to the other end of the Microphone Cable. Place the Radar Test Receiver a distance of 45 feet in front of the Chronograph and 4 feet above the ground. Connect the Cable Ass'y (stored in CARR) to the MICROPHONE input on the Chronograph and to the CHRONOGRAPH input connector on the CARR. Apply Primary Input Power per para. 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Level the Chronograph within $\pm$ 2 mils in both the horizontal and

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vertical planes. Place the Front Panel Monitor switch to the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2 Set the Toggle switch on the CARR Simulator to the METER position Adjust deflection and elevation of the Chronograph for a maximum meter indication on the CARR Simulator Monitor meter. Lock the Chronograph in the position of maximum meter indication. The vertical cross-hair in the Telescope XM128 shall be on the center of the Radar Test Receiver and the horizontal cross-hair shall be 10 inches above the center of the Radar Test Receiver. Set the Toggle switch on the CARR Simulator to the OFF position. Adjust the Leakage Tuning per para 4.6.2.5. Set the DELAY selector switch on the CARR Simulator to the CONT. position. Place the VELOCITY Selector switch on the CARR Simulator and the Chronograph to the 70-325 position. Set the MICROPHONE SENSITIVITY on the Chronograph to position 10. Place the Front Panel Monitor switch on the Chronograph to the SIGNAL position. Set the Toggle switch on the CARR Simulator to the OSC position. The Front Panel Monitor meter shall read in the Green Band. Place the DELAY selector on the Chronograph to position 1. Reset the Digital Counter. The Nixie Tube display shall read all zeros and the Reset light shall continuously light. Depress the START button on the CARR Simulator. The Nixie Tube Display shall read 228.0 as per the Decal in the cover of the CARR Simulator. Repeat the above test procedure by sequencing the VELOCITY Selector on the CARR Simulator and the Chronograph and the DELAY Selector on the Chronograph thru the remaining nine (9) positions. The Nixie Tube Dispaly shall be as given on the Decal in the CARR Simulator cover and the Front Panel Monitor meter shall read in the Green Band. Place the VELOCITY Selector on the CARR Simulator and on the Chronograph in the 70-325 position. Sequence the DELAY Selector on the CARR Simulator and on the Chronograph as given in Table V. The Nixie Tube Display shall be as given in Table V for each respective position of the DELAY Selector. Set the MICROPHONE SENSITIVITY on the Chronograph to position 5. Reset the Digital Counter. The Nixie Tube Display shall read all zeros and the Reset Light shall be continuously lighted. Depress the START button on the CARR Simulator. The Reset Light shall go out and the Nixie Tube Display shall read 228.0 with a tolerance as given in cover decal.

TABLE V

CHRONOGRAPH DELAY	CARR DELAY	DIGITAL INDICATOR NIXIE TUBE DISPLAY
1	1	228.0
2	1	INCORRECT
2	2	228.0
2	3	INCORRECT
3	3	228.0
4	3	INCORRECT
4	4	228.0
4	5	INCORRECT
5	5	228.0
6	5	INCORRECT
6	6	228.0
6	7	INCORRECT
7	7	228.0
8	7	INCORRECT

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TABLE V (Cont'd)

CHRONOGRAPH DELAY	CARR DELAY	DIGITAL INDICATOR NIXIE TUBE DISPLAY
8	8	228.0
8	9	INCORRECT
9	9	228.0
10	9	INCORRECT
10	10	228.0

4.6.2.15 Internal Calibration Signal - Connect a frequency counter and an oscilloscope to the Audio Output receptical TP A3J1. Place the VELOCITY Selector switch on the Audio Amplifier to the 925-1150 CAL position. Apply Primary Input Power per para 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor switch to the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2 and the Leakage Tuning per para 4.6.2.5. Depress tha RESET button on the Digital Counter. The Reset Light shall continuously light and the Nixie Tube Display shall read all zeros. Hold the OPER switch on the Digital Counter in the CAL 1 position. The Audio Output on TP A3J1 shall be an undistorted sine wave having a frequency of 75 KHz $\pm$.012% and an amplitude of 600 $\pm$ 300 millivolts peak-to-peak. The Front Panel Monitor meter shall read below the Green band (less than 60) with the Front Panel Monitor Selector switch in the AGC position. The Front Panel Monitor meter shall read in the Green Band (+60 to +80) with the Front Panel Monitor Selector switch in the SIGNAL position. Connect the Microphone thru the Microphone Cable to the MICROPHONE input on the Chronograph. Reset the Digital Counter. The Reset Light shall continuously light and the Nixie Tube Display shall read all zeros. Hold the OPER switch on the Digital Counter in the CAL 2 position. The Audio Output and the Front Panel Monitor meter readings shall be as given above. The Nixie Tube Display shall read 1070.7 $^{+0.3}_{-0.4}$.

4.6.2.16 Microphone - Connect the Microphone thru the Microphone Cable to the MICROPHONE input on the Chronograph. Place the VELOCITY Selector switch on the Audio Amplifier to the 925-1150 CAL position. Apply Primary Input Power per para 4.6.2.1 (a); then allow a five (5) minute warm-up interval prior to proceeding with the tests. Place the Front Panel Monitor switch to the B+ position. The Front Panel Monitor meter shall read on the Red Line in the middle of the Green Band. Adjust the Automatic Frequency Control (AFC) per para 4.6.2.4.2 and the Leakage Tuning per para 4.6.2.5. Depress the RESET button on the Digital Counter. The Reset Light shall continuously light and the Nixie Tube Display shall read all zeros. Place the MICROPHONE SENSITIVITY switch to the 5 position. Place the VELOCITY Selector in the 925-1150 CAL position. Hold the OPER switch in CAL 1. Lightly tap the Microphone. The Reset Light shall go out and the Nixie Tube Display shall read 1070.7 $^{+0.3}_{-0.4}$.

4.6.2.16.1 Repeat the above test procedure with the MICROPHONE SENSITIVITY switch set to the 10 position.

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4.6.2.17 Elasped Time Meter - Record the time indication of the ELASPED TIME METER. Apply Primary Input Power per para 4.6.2.1 (a) and record the time of application. The ELAPSED TIME meter shall have an accuracy of ± 10% with a minimum of one (1) hour of application of Primary Input Power.

4.6.2.17.1 Repeat the above test procedure with an applied Primary Input Power per para 4.6.2.1 (b) and per para 4.6.2.1 (c).

4.6.3 Reliability Run-in - The following Run-in tests shall be performed on a 100% basis.

4.6.3.1 Temperature Run-in - The Chronograph shall be placed in a test chamber with an Equipment Configuration as defined in para 4.6.1.1. Without Primary Input Power applied, the temperature shall be gradually raised to a temperature of plus 125°F $\pm^{5°F}_{0°F}$ at a rate of not to exceed 2°F per hour. After temperature stabilization, apply Primary Input Power as defined in the respective test paragraphs undergoing test. Allow a five (5) minute warm-up interval prior to performing the first test. The time duration of this test at each temperature extreme shall be 3 hours $\pm^{15 \text{ minutes}}_{0 \text{ minutes}}$. During and subsequent to the Temperature Run-in, the tests to be performed shall be as defined in para 3.7.1. At all other times during a temperature extreme when a specific test (para 4.6.2) is not being performed, Primary Input Power shall be applied as given in para 4.6.2.1 (a), with conditions as specified in para 3.7.1. Failures occurring during or subsequent to the Temperature Run-in shall be corrected as per para 3.7.4.

4.6.3.1.1. Repeat the above test procedure (para 4.6.3.1) at a temperature of minus 25°F $\pm^{5°F}_{0°F}$. Except for the change in temperature extreme from +125°F to -25°F, all other test conditions shall be as given in para 4.6.3.1. Failures occurring during or subsequent to a Temperature Run-in shall be corrected as per para 3.7.4

4.6.3.2 Vibration Run-in - The Vibration Run-in shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.7.2. The Equipment Configuration shall be as defined in para 4.6.1.1. At all other times during the vibration when a specified test (para 4.6.2) is not being performed, Primary Input Power shall be applied as given in para 4.6.2.1 (a) with conditions as specified in para 3.7.2. Failures occurring during or subsequent to Vibration Run-in shall be corrected per para 3.7.4.

4.6.3.3 Shock Run-in - The Shock Run-in shall be performed as specified in MIL-STD-810, Method 516 at the conditions specified in 3.7.3. The Equipment Configuration shall be as defined in para 4.6.1.1. During the shock, Primary Input Power shall be applied as given in para 4.6.2.1 (a) with conditions as specified in para 3.7.3. Failures occurring during or subsequent to Shock Run-in shall be corrected per para 3.7.4.

4.6.3.4 Failures - Failures during Reliability Run-in shall not be cause for rejection, but shall be corrected as defined in para 3.7.4.

4.6.4 Packaging, Packing and Marking - Quality Assurance Provisions for packaging, packing and marking shall be as specified in MIL-P-14232 and Section 5 herein.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with Packing Data Sheet 10525235. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended Use - The Chronograph is used as a System Component of a Fire Direction System to obtain projectile velocities.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, Number and the date of this specification.

(b) Selection of applicable levels of preservation, packing and packing.

(c) Applicable packaging data sheet (see 5.1)

(d) Applicable stock number

(e) Provisions for First Article Testing

6.3 Definitions - The term Chronograph referenced in this specification refers to the complete Chronograph Set including each component, sub-assembly, part, and accessory which comprises the Set.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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